Statistical Power and Sample Size Concepts

Read the following case study to reinforce your understanding of the sampling concepts discussed in Session 5.

Researchers are evaluating whether a new interactive teaching method improves students' math test scores compared to traditional teaching methods. The math scores are measured on a 100-point standardised test.

- The **primary outcome** of this study is the end-of-term standardised math test score out of 100 points
- If the researchers expect the new method to increase scores by 5 points and the standard deviation of scores is known to be 10, the **effect size** is calculated using Cohen's d:

$$\text{Effect Size (d)} = \frac{\text{Mean Difference}}{\text{Standard Deviation}}$$

This will give an effect size of 5/10 = 0.5 (medium)

- A medium effect size (d = 0.5) reflects a noticeable difference between groups and is easier to detect than smaller effects, requiring fewer participants for the same power. **Detecting smaller effects reliably requires larger sample sizes**
- If the researchers want 80% statistical power to detect this effect size at a significance level of α = 0.05, **the number of students needed in each group** can be calculated using Cohen's conventional sample size table.
- **Cohen's conventional sample size table** for two independent means is used to determine the sample size required per group to detect a specific effect size (d) with a given power (usually 0.80) and significance level (usually α = 0.05, two-tailed).

Effect Size (d)	Power (1-β)	α = 0.05 (two-tailed)	Sample Size per Group
0.2 (small)	0.80	0.05	394
0.5 (medium)	0.80	0.05	**64**
0.8 (large)	0.80	0.05	26

Effect Size (d): The standardised difference between means.

- Small = 0.20
- Medium = 0.50
- Large = 0.80

Power (1 - β): Probability of detecting an effect if it exists (commonly 0.80 or 0.90).
Alpha (α): Probability of Type I error (commonly set at 0.05, two-tailed).
If a medium effect size (d = 0.50) is expected using an 80% power, 64 participants would be needed per group.

- If researchers double sample size per group, the **statistical power of the study will increase significantly**; possibly to over 95%. **This means the study is more likely to detect a true effect if one exists**.
- **Lowering α (the significance level) reduces the chance of Type I error** (false positives), making the test more stringent
- Although **lowering α to 0.01** reduces the risk of a Type I error (false positive) it **reduces statistical power**. There is a trade-off to maintain the same power (e.g., 80%) and that is to use a larger sample size
- If the student performance is **highly variable** due to different teaching quality and socioeconomic backgrounds for example, this **variability will impact the required sample size**. High variability increases the "noise" in the data, making it harder to detect true differences between groups. As a result, a **larger sample size will be needed to maintain statistical power**.
- If the new teaching method actually works but the study finds no statistically significant difference in scores, this would be **failing to detect a real effect** (false negative) and could possibly be due to **low power**, **small sample size**, or **high variability**.

Key take aways:

- Larger sample sizes increase power
- Larger effect sizes are easier to detect, requiring smaller samples
- Lower α levels reduce Type I error but require more data to maintain power
- Higher variability increases the required sample size to detect an effect

All four factors interact to determine whether the study can reliably detect the true impact of the teaching method.